

1. Bob draws cards (with replacement, and with shuffles in between each draw) from a deck of 52 cards, until he gets an Ace, and then he stops.

Let X denote the number of draws until the first Ace appears (including the draw for the Ace itself).

1a. For each $x \geq 1$, find $P(X = x)$.

1b. For each $x \geq 1$, find $P(X \geq x)$.

2. Flip 100 (fair) coins in a row. Let X denote the number of heads that appear. For each x with $0 \leq x \leq 100$, find $P(X = x)$.

3. Consider a randomly chosen bear family, consisting of 3 bears, namely, a daddy bear, a momma bear, and a baby bear. Each bear has six possible colors, namely, red, orange, yellow, green, blue, purple. If we pick such a bear family randomly, the sample space consists of all ordered 3-tuples whose elements are r, o, y, g, b, p . Assume that all such outcomes are equally likely. Let X denote the number of green bears in a randomly selected bear family. For $0 \leq x \leq 3$, find $P(X = x)$.

4. Suppose that a drawer of t-shirts contains 7 t-shirts from Purdue and 3 t-shirts from IU. Alice randomly selects t-shirts from the drawer, **without replacement**, until the first IU t-shirt is selected. All possible orders of selection are equally likely. Let X denote the number of t-shirts she chooses, until the first IU t-shirt appears (including the draw for the IU t-shirt itself).

For $1 \leq x \leq 8$, find $P(X = x)$.

5. Alice rolls a 6-sided die. Whatever value she gets, Bob tosses that many fair coins. (For example, if Alice rolls a “5”, then Bob tosses 5 fair coins.)

Suppose that Bob gets exactly 1 head. (This is a little different than our question last week, in which we supposed that Bob gets 0 heads.)

Let Y denote the value on Alice’s die. Let X denote the number of heads that Bob has.

5a. What is $P(Y = 1 \mid X = 1)$?

5b. What is $P(Y = 2 \mid X = 1)$?

5c. What is $P(Y = 3 \mid X = 1)$?

5d. What is $P(Y = 4 \mid X = 1)$?

5e. What is $P(Y = 5 \mid X = 1)$?

5f. What is $P(Y = 6 \mid X = 1)$?